

Perception Module (Section 3.3)

Camera



AprilTag



LAB State



LLM Module (Section 3.2)

“Move bottle0 to tray0,
transfer the contents of beaker0 into flask0,
and then stir flask0 for 30s.”

XDL (Chemical Description Language)

XDL
Validator

```
<Procedure>
<Move object="bottle0" place="tray0" />
<Transfer from_vessel="beaker0" to_vessel="flask0" />
<Stir vessel="flask0" time="30" />
</Procedure>
```

World State

Current Procedure
Next Procedure

Tool Selection

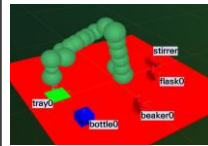


Rearrangement



TAMP Module (Section 3.4)

World State
Initializer



Goal State

```
HandEmpty()
Poured("beaker0",
"flask0", "water")
```

Symbolic
Task Plan

Pick

MoveHolding

⋮

Batched
Optimization Solver

Constrained
Sampler

Parameter
Optimizer

Motion
Solver

Motion Trajectory



Skills Library Module (Section 3.5)

Manipulation

Pouring

Device
Operation

Stirring

Tool
Changing

Execution

